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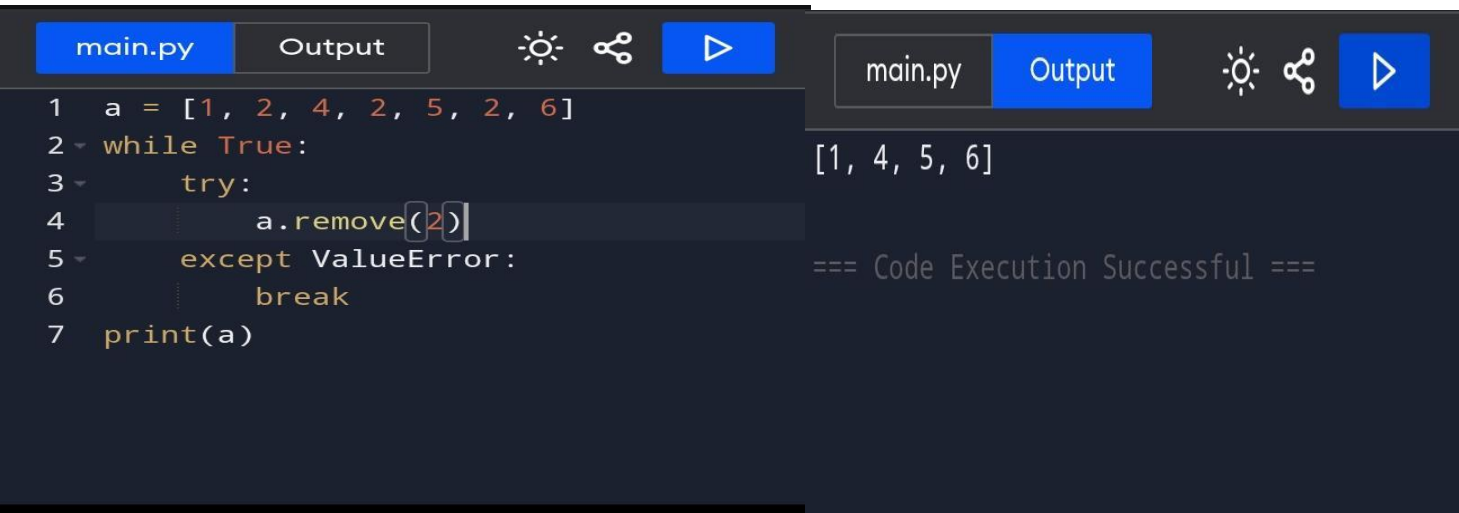
ID:FA25-BMD-D-015

Subject: Programming Fundamentals

LISTS

Lab programs M1 till M10

M1

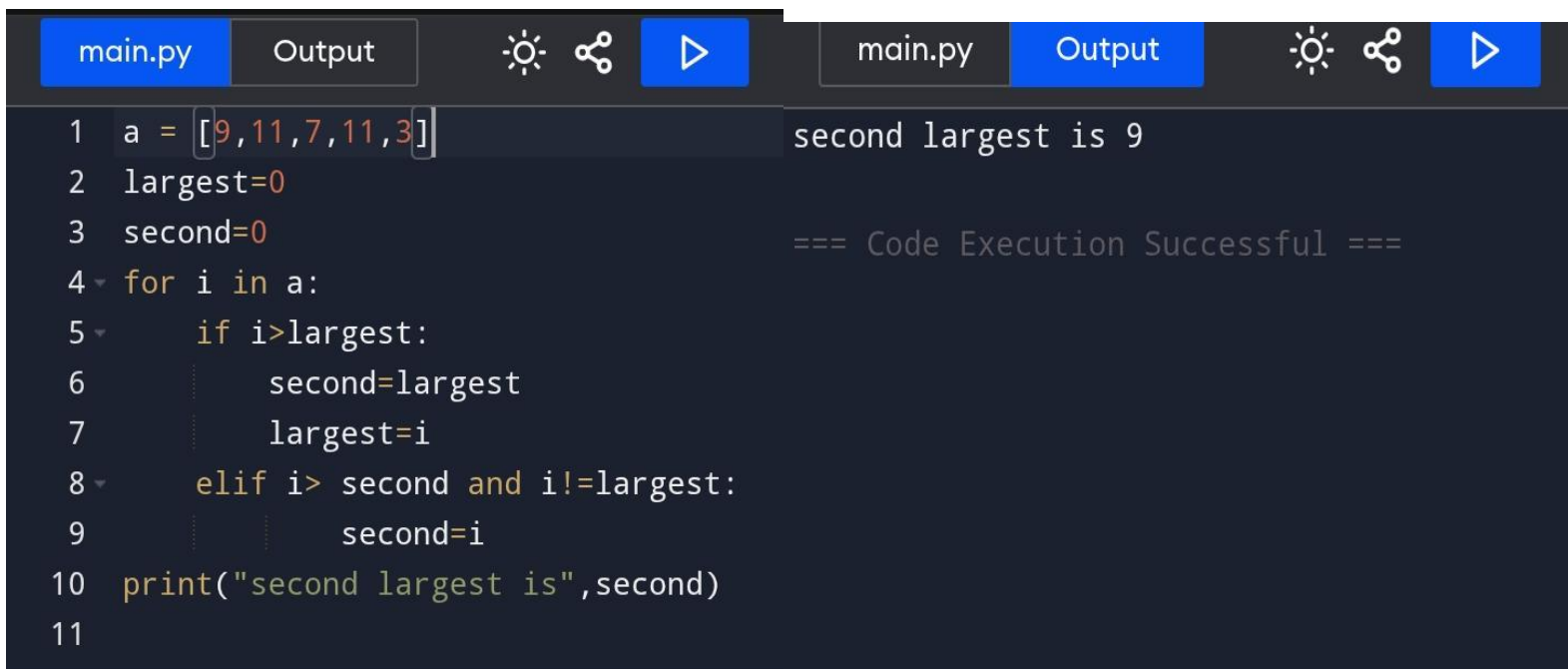


```
main.py Output [1, 4, 5, 6]
```

```
1 a = [1, 2, 4, 2, 5, 2, 6]
2 while True:
3     try:
4         a.remove(2)
5     except ValueError:
6         break
7 print(a)
```

=== Code Execution Successful ===

M2



```
main.py Output second largest is 9
```

```
1 a = [9, 11, 7, 11, 3]
2 largest=0
3 second=0
4 for i in a:
5     if i>largest:
6         second=largest
7         largest=i
8     elif i> second and i!=largest:
9         second=i
10 print("second largest is",second)
11
```

=== Code Execution Successful ===

M3

Programiz

Python Online Compiler

Programiz PRO

main.py

Output



```
1 a = [10, 20, 30, 40, 50]
2 a.append(a.pop(0))
3 print(a)
```

Programiz

Python Online Compiler

Programiz PRO

main.py

Output



```
[20, 30, 40, 50, 10]
```

=== Code Execution Successful ===

M4

Programiz

Python Online Compiler

Programiz PRO

main.py

Output



```
1 a = [1, 2, 3, 4, 5]
2 k = 2
3 value = 99
4 a = a[:k] + [value] + a[k:]
5 print(a)
```

Programiz

Python Online Compiler

Programiz PRO

main.py

Output



```
[1, 2, 99, 3, 4, 5]
```

=== Code Execution Successful ===

M5

main.py	Output
<pre>1 a = [5, 6, 7, 8, 9, 10] 2 k = 0 3 for i in range(0, 5): 4 a[i] = a[i + 1] 5 a.pop() 6 print(a)</pre>	<pre>[6, 7, 8, 9, 10] === Code Execution Successful ===</pre>

M6

Programiz Python Online Compiler	Programiz PRO				
<table border="1"><thead><tr><th>main.py</th><th>Output</th></tr></thead><tbody><tr><td><pre>1 line = input("Enter a line: ") 2 words = line.split() 3 longest = max(words, key=len) 4 print("Longest:", longest) 5 print("Length:", len(longest))</pre></td><td><pre>Enter a line: Apple is a fruit Longest: Apple Length: 5 === Code Execution Successful ===</pre></td></tr></tbody></table>	main.py	Output	<pre>1 line = input("Enter a line: ") 2 words = line.split() 3 longest = max(words, key=len) 4 print("Longest:", longest) 5 print("Length:", len(longest))</pre>	<pre>Enter a line: Apple is a fruit Longest: Apple Length: 5 === Code Execution Successful ===</pre>	
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<pre>1 line = input("Enter a line: ") 2 words = line.split() 3 longest = max(words, key=len) 4 print("Longest:", longest) 5 print("Length:", len(longest))</pre>	<pre>Enter a line: Apple is a fruit Longest: Apple Length: 5 === Code Execution Successful ===</pre>				

M7

```
main.py Output
1 s = input("Enter sentence: ")
2 words = s.split()
3 result = words[0]
4 for i in range(1, len(words)):
5     result = result + " | " +
        words[i]
6 print(result)
```

```
main.py Output
Enter sentence: She is eating her lunch
She | is | eating | her | lunch

=== Code Execution Successful ===
```

M8

```
Python Online Compiler Programiz PRO
main.py Output
1 a = [3, 3, 2, 3, 1, 2, 4, 4]
2 u = {}
3 for x in a:
4     u[x] = None
5 print(list(u))
```

```
Programiz
Python Online Compiler Programiz PRO
main.py Output
[3, 2, 1, 4]

=== Code Execution Successful ===
```

M

main.pyOutput



```
1 a = [1, 1, 1, 2, 2, 3, 4, 4]
2 done = []
3 for x in a:
4     if x not in done:
5         print(x, "->", a.count(x))
6         done.append(x)
```

main.pyOutput



```
1 -> 3
2 -> 2
3 -> 1
4 -> 2

=== Code Execution Successful ===
```

M10

main.pyOutput



```
1 a = [8, 3, 5, 2, 9, 1]
2 for _ in range(len(a)):
3     for i in range(len(a) - 1):
4         if a[i] > a[i + 1]:
5             a[i], a[i + 1] = a[i +
6                 1], a[i]
7 print(a)
```

main.pyOutput



```
[1, 2, 3, 5, 8, 9]

=== Code Execution Successful ===
```